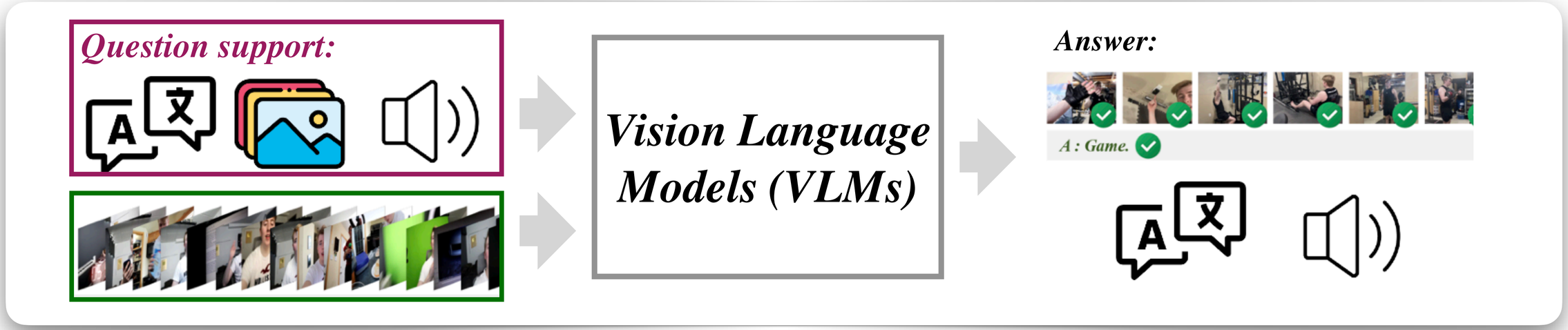
Plan 4: Human-level Visual Intelligence



Key challenges

A. More Inteligent interaction

- *Language & visual & audio*
- *Motion-sensitive understanding*
- *Long-term causal reasoning*

B. Longer videos: 1-min, 15-min, 60-min, 2-hour, 2-day, 2-month

- *Limited Capacity (Model)*

NSF, CISE, Future CoRe

Robust Intelligence (RI)

- Small
- $500,000
- 3-year support

Faculty Early Career Development Program (CAREER)

- July, 2029-2031
- 5-year support

Expected Outcomes (3–5 Years)

Teaching

- **High-quality teaching**
  - Undergraduate courses
  - Graduate courses
- **Develop new courses**
  - AI on engineering problems
  - Cutting-edge technologies
- **Scientific activities**
  - Interdisciplinary seminars
  - Science Outreach

Research Team & Goals

- **Build my own lab**
  - A small but impact team
  - Recruit & train students
- **Keep a research record**
  - Papers on top conferences
  - Organize AI seminars
- **Academic services**
  - AC on AI/ML conferences
  - IEEE / ACM

Funding! Funding!!

- **Year 1-2 (Plan 1)**
  - **NSF Core (CISE / IIS)**
  - **ONR & AFOSR**
  - **Industry grants**
- **Year 3-5 (Plan 1 & 2)**
  - **NSF CAREER Award**
  - **DARPA & NASA**
  - **Industry grants**